HW Set #4 CHE-311

1.) Navier-Stokes equation

$$\rho\left(\frac{\partial v_x}{\partial t}+v_x\frac{\partial v_z}{\partial x}+v_z\frac{\partial v_z}{\partial z}\right)=-\frac{\partial p}{\partial z}+\mu\left(\frac{\partial^2 v_z}{\partial x^2}+\frac{\partial^2 v_z}{\partial z^2}\right)$$

- flow is unidirectional : $v_z = v_z(x,t)$ $\quad \frac{\partial v_x}{\partial z}=0 \therefore \frac{\partial^2 v_x}{\partial z^2}=0$
- $v_x = 0$

equation reduced to

$$\rho\frac{\partial v_x}{\partial t}=-\frac{\partial p}{\partial z}+\mu\frac{\partial^2 v_z}{\partial x^2}$$

Gradient is constant, $\frac{\partial p}{\partial z}=\frac{\Delta p}{L}$

$$\boxed{\rho\frac{\partial v_x}{\partial t}=\mu\frac{\partial^2 v_z}{\partial x^2}-\frac{\Delta p}{L}}$$

2.) SS $\frac{\partial v_z}{\partial t}=0$ $\quad \mu\frac{\partial^2 v_{z\infty}}{\partial x^2}-\frac{\Delta p}{L}=0 \rightarrow \frac{\partial^2 v_{z\infty}}{\partial x^2}=\frac{\Delta p}{\mu L}$

Integrate $\frac{\partial v_{z\infty}}{\partial x}=\frac{\Delta p}{\mu L}x+C_1$ integrate again $v_{z\infty}=\frac{\Delta p}{2\mu L}x^2+C_1x+C_2$

Apply symmetry condition $\left.\frac{\partial v_{z\infty}}{\partial x}\right|_{x=0}=0 \rightarrow \frac{\partial v_{z\infty}}{\partial x}=\frac{\Delta p}{\mu L}x+C_1$

at $x=0$ $\quad C_1=0$ $\quad v_{z\infty}=\frac{\Delta p}{2\mu L}x^2+C_2$ Using $x=B$ $\quad 0=\frac{\Delta p}{2\mu L}B^2+C_2 \rightarrow C_2=-\frac{\Delta p}{2\mu L}B^2$

$$v_{z\infty}=-\frac{B^2\Delta p}{2\mu L}\left[1-\left(\frac{x}{B}\right)^2\right] \xrightarrow[\text{Velocity profile}]{SS} \boxed{v_{z\infty}=v_{max}\left[1-\left(\frac{x}{B}\right)^2\right]} \rightarrow \boxed{v_{max}=\frac{B^2\Delta p}{2\mu L}}$$

3.) Sub $v_z = v_{max}\phi$ $\quad \frac{\partial v_z}{\partial t}=\frac{\partial z}{\partial t}\frac{\partial \phi}{\partial z}$ $\quad \tau=\frac{\mu t}{\rho B^2} \rightarrow \frac{\partial z}{\partial t}=\frac{\mu}{\rho B^2} \rightarrow \frac{\partial v_z}{\partial t}=v_{max}\frac{\mu}{\rho B^2}\frac{\partial \phi}{\partial z}$

$$\rho\frac{\partial v_z}{\partial t}=\frac{\mu v_{max}}{B^2}\frac{\partial \phi}{\partial z}$$

$\eta=\frac{x}{B} \rightarrow \frac{\partial \eta}{\partial x}=\frac{1}{B} \rightarrow \frac{\partial v_x}{\partial x}=v_{max}\frac{\partial \phi}{\partial \eta}\frac{\partial \eta}{\partial x}=v_{max}\frac{1}{B}\frac{\partial \phi}{\partial \eta}$ 2nd derivative $\frac{\partial^2 v_z}{\partial x^2}=v_{max}\frac{1}{B^2}\frac{\partial^2 \phi}{\partial \eta^2}$

sub both terms $\quad \rho\frac{\partial v_z}{\partial t}=\mu\frac{\partial^2 v_z}{\partial x^2}-\frac{\Delta p}{L} \rightarrow \frac{\mu v_{max}}{B^2}\frac{\partial \phi}{\partial z}=\mu\left(v_{max}\frac{1}{B^2}\frac{\partial^2 \phi}{\partial \eta^2}\right)-\frac{\Delta p}{L}$

divide by $\frac{\mu v_{max}}{B^2}$ $\quad \frac{\partial \phi}{\partial z}=\frac{\partial^2 \phi}{\partial \eta^2}-\frac{\Delta p}{L}\cdot\frac{B^2}{\mu v_{max}} \rightarrow \frac{B^2}{\mu v_{max}}\cdot\frac{\Delta p}{L}=2 \rightarrow \boxed{\frac{\partial \phi}{\partial z}=\frac{\partial^2 \phi}{\partial \eta^2}+2}$

4.) $\frac{\partial^2 \phi_s}{\partial \eta^2} + 2 = 0 \rightarrow \frac{\partial^2 \phi_s}{\partial \eta^2} = -2$ Integrate $\frac{\partial \phi_s}{\partial \eta} = -2\eta + C_1$

$\left.\frac{\partial \phi_s}{\partial \eta}\right|_{\eta=0} = 0 \xrightarrow{C_1 = 0} \frac{\partial \phi_s}{\partial \eta} = -2\eta$ Integrate $\phi_s = -\eta^2 + C_2$

no-slip condition $\eta = 1$ $\phi_s(1) = 0$; $0 = -1 + C_2 \rightarrow C_2 = 1$; $\boxed{\phi_s = 1 - \eta^2}$

5.) Walls are at $\eta = -1$ $\boxed{\phi(-1, \tau) = 0}$ $\eta = +1$ $\boxed{\phi(+1, \tau) = 0}$ $\tau_2 = 0$ at initial $t = 0$ condition $\boxed{\phi(\eta, 0) = 0}$

6.) Cosine expansion $\phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos\left(\frac{2n-1}{2} \pi \eta\right)$

Define $\lambda_n = \frac{2n-1}{2}\pi \rightarrow \phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos(\lambda_n \eta)$

$\frac{\partial \phi}{\partial \tau} = \sum_{n=1}^{\infty} \frac{\partial C_n}{\partial \tau} \cos(\lambda_n \eta)$ Time derivative

2nd derivative $\frac{\partial^2 \phi}{\partial \eta^2} = \sum_{n=1}^{\infty} C_n(\tau) \frac{\partial^2}{\partial \eta^2} \cos(\lambda_n \eta)$

$\frac{\partial^2}{\partial \eta^2} \cos(\lambda_n \eta) = -\lambda_n^2 \cos(\lambda_n \eta) \rightarrow \frac{\partial^2 \phi}{\partial \eta^2} = -\sum_{n=1}^{\infty} \lambda_n^2 C_n(\tau) \cos(\lambda_n \eta)$

$\sum_{n=1}^{\infty} \frac{\partial C_n}{\partial \tau} \cos(\lambda_n \eta) = -\sum_{n=1}^{\infty} \lambda_n^2 C_n(\tau) \cos(\lambda_n \eta) + 2$

$\sum_{n=1}^{\infty} \left(\frac{\partial C_n}{\partial \tau} + \lambda_n^2 C_n\right) \cos(\lambda_n \eta) = 2$

$2 = \sum_{n=1}^{\infty} b_n \cos(\lambda_n \eta) \rightarrow b_n \frac{\int_{-1}^{1} 2\cos(\lambda_n \eta)\, d\eta}{\int_{-1}^{1} \cos^2(\lambda_n \eta)\, d\eta}$ $\int_{-1}^{1} \cos^2(\lambda_n \eta)\, d\eta = 1$

$b_n = 2\int_{-1}^{1} \cos(\lambda_n \eta)\, d\eta$

$\int_{-1}^{1} \cos(\lambda_n \eta)\, d\eta = \left[\frac{\sin(\lambda_n \eta)}{\lambda_n}\right]_{-1}^{1} = \frac{2\sin(\lambda_n)}{\lambda_n} \rightarrow b_n = 2\left(\frac{2\sin(\lambda_n)}{\lambda_n}\right) = \frac{4\sin(\lambda_n)}{\lambda_n}$

$\lambda_n = \frac{2n-1}{2}\pi$ $\sin(\lambda_n) = \sin\left(\frac{2n-1}{2}\pi\right) = (-1)^{n-1} \rightarrow b_n = \frac{4(-1)^{n-1}}{\lambda_n} \rightarrow b_n = \frac{8(-1)^{n-1}}{(2n-1)\pi}$

$$2 = \sum_{n=1}^{\infty} \frac{8(-1)^{n-1}}{(2n-1)\pi} \cos(\lambda_n \eta)$$

ODE $\frac{\partial C_n}{\partial \tau} + \lambda_n^2 C_n = \frac{8(-1)^{n-1}}{(2n-1)\pi} \rightarrow \frac{\partial C_n}{\partial \tau} + \lambda_n^2 C_n = \frac{8(-1)^{n-1}}{(2n-1)\pi}$

for $C_n(\tau)$

Solve for ODE $\frac{\partial C_n}{\partial \tau} + \lambda_n^2 C_n = b_n \rightarrow C_n(\tau) = \frac{b_n}{\lambda_n^2} + K_n e^{-\lambda_n^2 \tau}$

$\lambda_n^2 = \left(\frac{2n-1}{2}\pi\right)^2 = \frac{(2n-1)^2\pi^2}{4} \rightarrow \frac{d_n}{\lambda_n^2} = \frac{8(-1)^{n-1}}{(2n-1)\pi} \cdot \frac{4}{(2n-1)^2\pi^2} = \frac{32(-1)^{n-1}}{(2n-1)^3\pi^3}$

$$C_n(\tau) = \frac{32(-1)^{n-1}}{(2n-1)^3\pi^3} + K_n e^{-\lambda_n^2 \tau}$$

Initial condition: At $\tau = 0$, $\phi(\eta, 0) = 0$

$$\sum_{n=1}^{\infty} C_n(0)\cos(\lambda_n \eta) = 0 \therefore 0 = \frac{32(-1)^{n-1}}{(2n-1)^3\pi^3} + K_n \rightarrow K_n = -\frac{32(-1)^{n-1}}{(2n-1)^3\pi^3}$$

Final coefficient

$$C_n(\tau) = \frac{32(-1)^{n-1}}{(2n-1)^3\pi^3}\left[1 - e^{-\lambda_n^2 \tau}\right] \xrightarrow[\lambda_n^2]{\text{replace}} C_n(\tau) = \frac{32(-1)^{n-1}}{(2n-1)^3\pi^3}\left[1 - \exp\left(-\frac{(2n-1)^2\pi^2}{4}\tau\right)\right]$$

$$\boxed{\phi(\eta, \tau) = \sum_{n=1}^{\infty} \frac{32(-1)^{n-1}}{(2n-1)^3\pi^3}\left[1 - \exp\left(-\frac{(2n-1)^2\pi^2}{4}\tau\right)\right]\cos\left(\frac{2n-1}{2}\pi\eta\right)}$$

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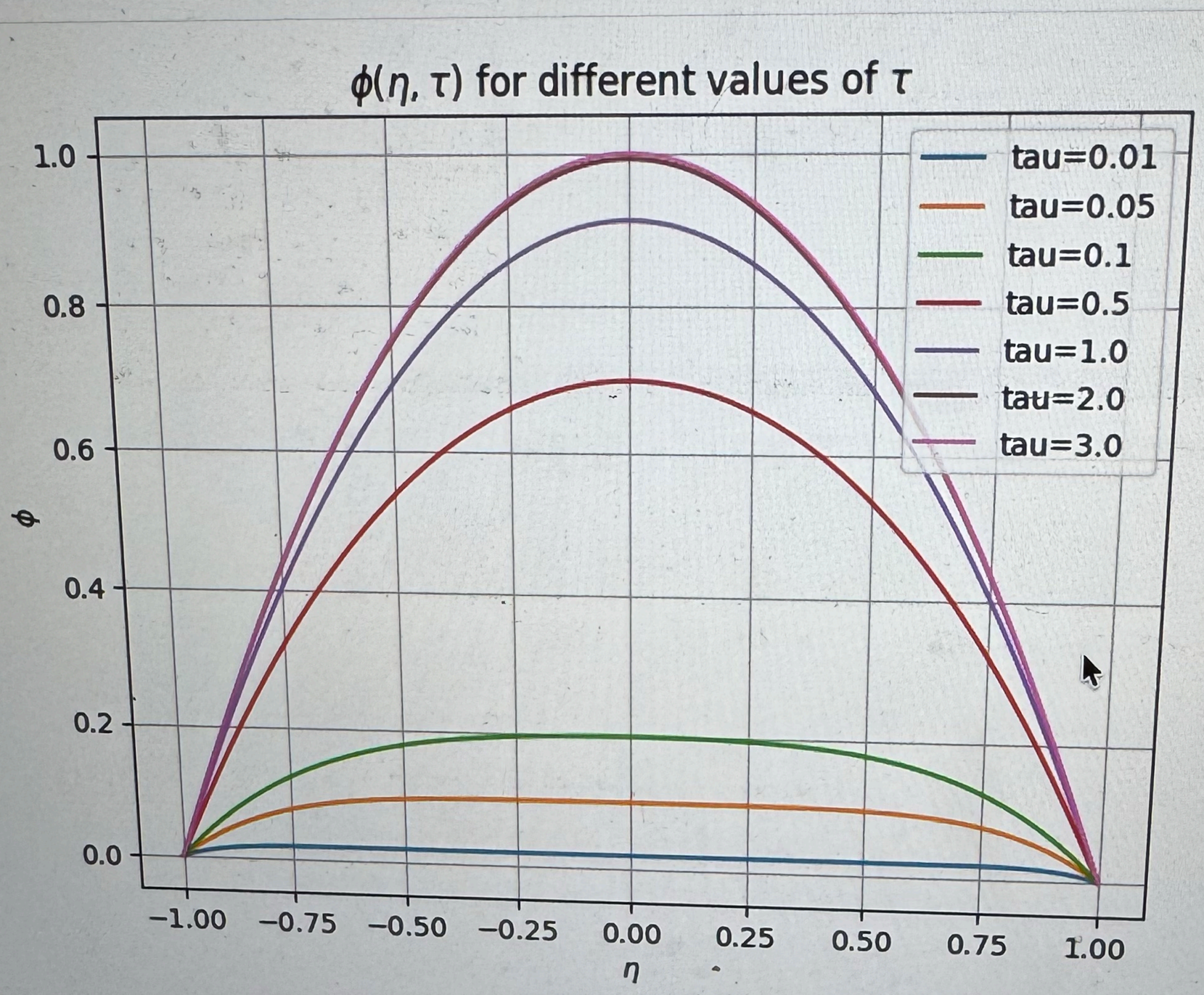

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